Simulation Concept Description

My project is a simulation called Six Day Garden. In Six Day Garden, the player is challenged to grow the mysterious Universe Flower so that it blooms in time for the Horticulture Fair Competition in 6 days. Nobody has successfully gotten this flower to bloom in years, as it is very fragile. The player who keeps the flower perfectly healthy will win 1st place at the Fair.

Actors

Universe Flower

The Universe Flower (UF) has three growth stages. The UF starts out at stage 1, a small clover. To get the UF to stage 2, the player must make it to day 3 with the UF still alive. The UF is alive as long as the plant health count is greater than 1. To get the UF to stage 3, the player must get to day 5 with perfect plant health.

Compost

There are six pieces of fruit that can be used as compost: an apple, a banana, orange, pear, blueberry, and strawberry.

Aphids

The aphids are small red bugs that may appear on the plant. They damage the plant if they aren't removed before the end of the day.

Action buttons (left to right)

- First, there's a blue button which waters the flower.
- The yellow button adds a piece of compost to the soil.
- The red button sprays pest spray onto the plant.
- The green button lets the player sleep until the next day.

Counters

- The day count begins at 1. This is displayed on the top left as text. This will increase up to 6 each time the player sleeps, provided that their plant is alive.
- The growth stage count begins at 1. This is displayed by the look and size of the plant.
- The plant health count begins at 6. This is displayed on the top left as text. This will never increase but it can decrease.
- The water count begins at 0. This is indicated by a blue dot on the plant. No blue dot indicates 0 for that day, and one blue dot indicates 1 or greater. Each day begins with a water count of 0.
- The pest count begins at 0. This is indicated by red aphids on the plant. On days 2-5, the plant has a 50% chance to either have 3-5 aphids or 0 aphids.
- The compost count begins at 0. This is indicated by individual fruits near the base of the plant. There is a maximum of 6 at once.

Github: https://github.com/madisonghi/CSC170